Bayesian Techniques in Macroeconomics

Academic year 2026–2027

Format:	Four sessions, 08:30–11:45
Course page:	https://vermandel.org/btm/
Public material:	https://github.com/Vermandel/bayesian-techniques-macroeconomics

Course description

This course develops the practical and conceptual tools used to estimate modern macroeconomic models. Starting from business-cycle measurement and simple rational-expectations economies, it introduces state-space methods, maximum likelihood and Bayesian estimation of DSGE models. The emphasis is on a transparent workflow: economic assumptions, data, likelihood, computation, diagnostics and interpretation. A fuller technical description of this workflow – from the model's state-space representation to the Kalman filter, the likelihood, and Bayesian posterior inference with Dynare – is given in `docs/course-description.md`.

Learning objectives

By the end of the course, students should be able to:

- formulate and solve a small macroeconomic model with uncertainty.
- translate a model into a state-space representation and explain filtering.
- construct and interpret a likelihood for macroeconomic data.
- estimate a DSGE model with MATLAB and Dynare using likelihood-based and Bayesian tools.
- document a reproducible data and estimation workflow, including diagnostics.

Prerequisites

Students should be comfortable with graduate macroeconomics, linear algebra, probability and basic econometrics. Familiarity with MATLAB is expected. Prior experience with Dynare is useful but not required. The course material provides the necessary entry points.

Schedule and session structure

Date	Session	Main topics
12 November 2026	1. Modern Business Cycle Theory	Measurement, consumption–saving, expectations and approximation.
19 November 2026	2. A Production Economy	RBC structure, linear rational expectations and Dynare programming.
26 November 2026	3. Maximum Likelihood Estimation	State-space models, Kalman filtering, likelihood and DSGE estimation.
3 December 2026	4. Bayesian Estimation of DSGE Models	Priors, posterior inference, MCMC, diagnostics and policy analysis.

Teaching material

The public repository contains a compiled handout and oral slide deck for each session, MATLAB scripts, Dynare `.mod` files, small assessment models and concise run instructions. Private LaTeX sources and local theme assets are not part of the student release. Students should work from a clean clone of the repository rather than from a copy containing generated outputs.

Software, data and DBnomics

The course code is validated with MATLAB R2025a and Dynare 7.0. A compatible Dynare 7.x release may also be used. The course scripts use DBnomics for reproducible macroeconomic data retrieval. No downloaded dataset is stored in the repository: run `data/download_data.m` (or a documented assessment-specific downloader) before running a model. Record the provider, dataset, series identifier, frequency, retrieval date and all transformations. General DBnomics guidance is maintained in the shared DBnomics teaching repository. Course-specific scripts remain in this repository.

Assessment

Students choose one of the four published assessment models. The submission has two parts: an empirical/macroeconomic report and the complete code needed to reproduce it.

Submission deadline: 15 January 2027.

- Write the report in English and submit it as a PDF produced from LaTeX.
- Submit the complete reproducible code. It must run from documented instructions.
- Explain data sources and transformations. Present and interpret estimation results.
- Include relevant diagnostics, such as convergence and posterior or likelihood checks.
- Do not submit generated caches, Dynare output or unnecessary downloaded data.

Detailed model choices, deliverables and public rules are in `assessment/README.md`.

Grading rubric

Criterion	Weight
Economic model, derivation and mapping to code	15%
Data choice, provenance, transformations and measurement equations	15%
Technical validity (steady state, residuals, determinacy, estimation)	15%
Priors and identification argument	15%
Posterior computation and MCMC diagnostics	15%
Economic interpretation, uncertainty and robustness	15%
Reproducibility, code quality and documentation	10%

References

- An, S. and F. Schorfheide (2007), "Bayesian Analysis of DSGE Models," *Econometric Reviews*.
- Canova, F. (2007), *Methods for Applied Macroeconomic Research*. Princeton University Press.
- Fernández-Villaverde, J. (2010), "The Econometrics of DSGE Models," *SERIEs*.
- Herbst, E. and F. Schorfheide (2015), *Bayesian Estimation of DSGE Models*. Princeton University Press.
- Judd, K. (1998), *Numerical Methods in Economics*. MIT Press.

Contact and public links

Consult the course webpage for the current public information and the GitHub repository for materials and issue reporting related to reproducibility.